

-40 to 95°C and operating temperature of -30 to 85°C.

From a design engineer's perspective, what are the top factors—besides the obvious ones like price, size, brand, after-support, etc—that should be borne in mind while selecting the right LCD panel?

There are a few LCD specs the design engineers need to consider at high priority when they select the LCD. The first specification will be the screen size and aspect ratio. The aspect ratio is the ratio between the length and width of the LCD. Some common ratios are 4:3, 5:4, 16:9, and so on. Of course, sometimes marketing people will also consider these specs as they will affect the whole outlook and design of the product.

Then the engineer may need to consider the LCD's resolution and interface, whether they are matching with the motherboard. If the product is a semi-outdoor or outdoor application, then the engineer needs to also check the LCD's brightness and operating temperature range, because these are very important specs if the product is located in the sunshine.

Design engineers can also refer to the box item titled 'Selection Parameters for LCD Screens for Automotive Applications' (see previous two pages).

What are the common myths or missing info related to designing LCD screens for automotive solutions?

The smartphone has become very popular in recent years and it is influencing the engineers' design. We saw some EV companies use the smartphone LCD as the cluster or GPS display for their first-generation products. The smartphone LCD is nice but, unfortunately, it is not designed for automotive applications, especially not for 2-wheeler outdoor usage. When the 2-wheeler is under the sunshine, the driver



A head-up display (HUD) is any transparent display that presents data (Credit: GoMechanic)

can barely see anything from the smartphone LCD. And, also, the smartphone LCD's lifetime becomes much shorter under the automotive application scenario.

How do you support customers, design engineers, and R&D teams during their initial stages of identifying the right LCD for their design?

Sharp Singapore has been in this region for many years. We understand our customers. First, our team will get the customer's requirements from both the marketing and engineering sides. We will check the customer's motherboard's graphics capability, display interface, and other necessary technical details. We will propose the best suitable LCDs to the customer and explain the reason. We will explain what we observe from the market trend and help the customer to know the best options.

LCD samples and demo kits are available for the engineers to see the actual performance. There is also technical support available to help the design engineers to evaluate the LCD and design-in the LCD.

Do you have some form of sampling programme for them to receive samples during their prototyping

stages? Do you have development or evaluation kits for your LCD displays?

Sharp Singapore understands that samples and evaluation kits are important in the project's early stage. Evaluation kits are available for the engineer to evaluate the LCD performance during the proof of concept stage. Then we will provide sample LCDs for the customer's prototype builds.

What is your support structure for design engineers in India?

We have salespersons stationed in India at New Delhi and Bangalore. They are working closely with the customers' design engineers. There are technical support persons in Singapore and Japan. Our Indian team can support the customer onsite and bridge as technical person effectively between India and Singapore.

Are we correct to assume that you supply the displays through major distributors or through your India office?

We have many distributors in India who can support the customer's needs. Major OEMs are supported directly through our headquarters in Singapore. **EFY**